

2.2. Calculation and Verification of Frequency Deviation and Modulation Index of FM Signals Using MATLAB Simulation

PROGRAM (MATLAB CODE)

% Calculation and Verification of Frequency Deviation

% and Modulation Index of FM Signals

clc;

clear;

close all;

%% Input Parameters

Am = 2; % Message signal amplitude

fm = 500; % Message frequency (Hz)

Ac = 1; % Carrier amplitude

fc = 10000; % Carrier frequency (Hz)

kf = 1000; % Frequency sensitivity (Hz/V)

fs = 100000; % Sampling frequency (Hz)

T = 0.02; % Simulation duration (s)

t = 0:1/fs:T;

%% Message Signal

m = Am*cos(2*pi*fm*t);

%% Frequency Deviation Calculation

delta_f = kf * Am;

%% Modulation Index Calculation

```

beta = delta_f/fm;

%% FM Signal Generation

fm_signal = Ac*cos(2*pi*fc*t + beta*sin(2*pi*fm*t));

%% Plot Message Signal

figure;

subplot(2,1,1);

plot(t,m,'LineWidth',1.5);

title('Message Signal');

xlabel('Time (s)');

ylabel('Amplitude');

grid on;

%% Plot FM Signal

subplot(2,1,2);

plot(t,fm_signal,'LineWidth',1.5);

title('Frequency Modulated Signal');

xlabel('Time (s)');

ylabel('Amplitude');

grid on;

%% Spectrum Analysis

N = length(fm_signal);

FM_fft = fftshift(abs(fft(fm_signal))/N);

f = (-N/2:N/2-1)*(fs/N);

figure;

plot(f,FM_fft,'LineWidth',1.5);

title('FM Signal Spectrum');

```

```

xlabel('Frequency (Hz)');
ylabel('Magnitude');
grid on;
xlim([fc-10000 fc+10000]);

%% Display Results

fprintf('
FM SIGNAL PARAMETERS
');
fprintf('-----
');

fprintf('Carrier Frequency = %.0f Hz\n',fc);
fprintf('Message Frequency = %.0f Hz\n',fm);
fprintf('Message Amplitude = %.2f V\n',Am);
fprintf('Frequency Sensitivity = %.0f Hz/V\n',kf);
fprintf('
CALCULATED VALUES
');
fprintf('-----
');

fprintf('Frequency Deviation ( $\Delta f$ ) = %.0f Hz\n',delta_f);
fprintf('Modulation Index ( $\beta$ ) = %.2f\n',beta);

%% Carson's Rule Bandwidth

BW = 2*(delta_f + fm);

fprintf('Bandwidth (Carson Rule) = %.0f Hz\n',BW);

```

Procedure

1. Open MATLAB and create a new script file.
2. Clear the command window, workspace variables, and close all existing figure windows

using:

```
clc;
```

Clear;

close all;

3. Define the message signal parameters:

- Message amplitude $A_m=2V$

- Message frequency $f_m=500Hz$

4. Define the carrier signal parameters:

- Carrier amplitude $A_c=1V$

- Carrier frequency $f_c=10000Hz$

5. Set the frequency sensitivity constant $k_f=1000Hz/V$, sampling frequency $f_s=100000Hz$, and simulation duration $T=0.02s$.

6. Generate the time vector using: $t = 0:1/f_s:T$;

7. Generate the message signal using: $m = A_m \cos(2\pi f_m t)$;

8. Calculate the frequency deviation of the FM signal using: $\Delta f = k_f A_m$

9. Calculate the modulation index using: $\beta = \Delta f / f_m$

10. Generate the FM signal using: $fm_signal = A_c \cos(2\pi f_c t + \beta \sin(2\pi f_m t))$;

11. Plot the message signal in the time domain and observe its waveform.

12. Plot the generated FM signal and observe that the carrier frequency varies according to the message signal while the amplitude remains constant.

13. Determine the length of the FM signal: $N = \text{length}(fm_signal)$;

14. Compute the Fast Fourier Transform (FFT) of the FM signal: $FM_fft = \text{fft}(fm_signal)$;

15. Shift the spectrum to center the zero-frequency component and normalize it: $FM_fft = \text{fftshift}(\text{abs}(FM_fft)/N)$;

16. Generate the frequency axis corresponding to the FFT output: $f = (-N/2:N/2-1)*(fs/N)$;
17. Plot the frequency spectrum of the FM signal and observe the carrier component and sidebands.
18. Restrict the frequency axis around the carrier frequency for better visualization: `xlim([fc-10000 fc+10000]);`
19. Calculate the bandwidth of the FM signal using Carson's Rule:
 $BW = 2(\Delta f + f_m)$
20. Display the carrier frequency, message frequency, frequency deviation, modulation index, and bandwidth in the MATLAB command window.
21. Execute the program and verify that the calculated frequency deviation and modulation index match the values used in generating the FM **signal**

OUTPUT;

```
FMSIGNALPARAMETERS
-----
CarrierFrequency = 10000 Hz
MessageFrequency = 500 Hz
MessageAmplitude = 2.00 V
FrequencySensitivity = 1000 Hz/V
CALCULATEDVALUES
-----
FrequencyDeviation( $\Delta f$ )= 2000 Hz
ModulationIndex( $\beta$ )= 4.00
Bandwidth (CarsonRule) = 5000 Hz
```

OUTPUT (GRAPH)

